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LARGE LANGUAGE MODEL-BASED EMOTIONAL SPEECH ANNOTATION USING CONTEXT AND ACOUSTIC FEATURE FOR SPEECH EMOTION RECOGNITION

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ABSTRACT

The remarkable emergence of large language models (LLM) and their vast capabilities have opened a possibility for applications in various fields, including speech emotion recognition (SER). Despite the advancement of SER methods and the abundance of speech data, the requirement of having speech data labeled with emotions is a challenge to fulfill, considering the cost of human annotation. In this study, we propose utilizing LLM to annotate emotional speeches, investigating the use of conversation sequence transcription, and incorporating the textual acoustic feature descriptors into the prompt. Furthermore, we also examine the application of annotation results on emotional speeches as training and augmentation data. Our experiment using the IEMOCAP dataset shows that emotional speech annotation using LLMs can outperform human annotation with possibly lower annotation costs. The SER trained using the annotation result as a whole training data or augmentation data reaches a performance close to state-of-the-art SER methods.

Index Terms— speech emotion recognition, large language model, data annotation, data augmentation

1. INTRODUCTION

In recent years, there has been a remarkable emergence of large language models (LLM) [1], advanced AI systems capable of understanding and generating human-like text. These models have revolutionized the field of natural language processing (NLP) through their ability to comprehend context, semantics, and nuances in text, resulting in various applications, including dialog summarization [2] and question-answering [3]. Both LLM applications show capability in extracting meaningful context and understanding different sentiments and writing styles. These encourage the exploration of LLM uses in various tasks in other domains, such as sentiment analysis, low-resource language learning, classification, and affective computing.

One of the most prominent tasks in affective computing is speech emotion recognition (SER), which focuses on extracting and interpreting emotional cues from spoken language. By analyzing various acoustic features like pitch, tone, and intensity, as well as linguistic patterns and context, sophisticated algorithms have been developed to discern the underlying emotions conveyed in speech. Many of the recent SER methods with state-of-the-art performance utilize deep learning-based methods such as attention networks [4–10]. Some others consider utilizing speech data using pretrained self-supervised speech model such as Wav2Vec 2.0 [11] and HuBERT [12], while others incorporate the fusion of acoustic and text features [6–9, 13] which further improves the SER performance. Some studies also investigate the combination of acoustic representations to improve SER performance [4, 14]. The result from high-performing SER holds immense promise for applications from improving human-computer interaction to enhancing mental health diagnostics.

Developing a high-performing SER requires a vast amount of labeled emotional speech data. Although there might be abundant speech data containing emotions, most of these are unlabeled. Currently, the most common way to obtain the emotion label of speech data is to conduct human annotation. Human annotation is a process with many notable challenges. First, as the nature of emotion is relative to the perceiver, human annotation might result in high subjectivity. The subjectivity can be reduced by having multiple annotators and deciding the emotion through a majority vote. This requirement to reach an agreement may increase the second problem: the expensive labor cost for annotators. These two problems can be solved by utilizing LLMs to annotate emotional speeches. LLMs are trained on a vast amount of texts and, therefore, can be highly generalized and might be suitable for data annotation based on text [15, 16]. Some recent studies have investigated LLMs to infer emotion categories from speech content such as transcribed text and use them for weak labels for SER [17]. Other studies have attempted to recognize emotional qualities in music from lyrics using LLMs [18]. However, up to date, studies on using LLMs to annotate emotional speech data while incorporating the information from acoustic features are still limited.

In this study, we investigate the annotation of emotional speech data using LLMs and incorporate the annotation result to evaluate the performance of SER. Here, we show that LLMs have enough performance to annotate emotional speech by prompting the transcriptions. However, speeches have acoustic features, which contain rich information to decide emotions and should be considered whenever available. We propose a simple procedure to provide conversation context and acoustic feature descriptors in LLMs. The idea of adding textual description is adopted from a related work, where the acoustic feature contained in an utterance is included as text information for SER [19]. In addition, we conduct SER, comparing the performance using annotation results and the original labels from the IEMOCAP dataset [20]. This investigation provides insight into using LLMs in annotating emotional speech data with fewer human resources.

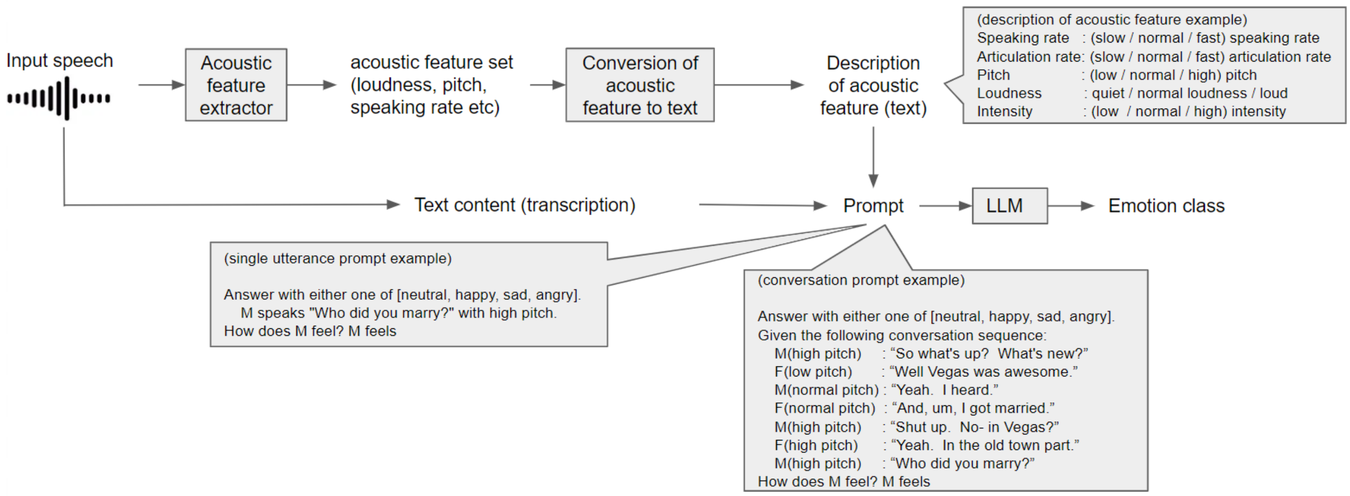

Fig. 1. Flow of annotation

The contribution of this paper is as follows:

- We presented an annotation method using LLM, incorporating the information from the conversation context and acoustic feature descriptors.
- We investigated the use of annotation results as augmented data for SER, and the performance is shown to be close to that of the state-of-the-art method in the IEMOCAP dataset.

2. METHODOLOGY

2.1. Overview

Our proposed method is divided into two steps. First, we conduct annotation of emotional speech data using LLM. Second, we perform the SER classification using some or all of the data from the annotation experiment, obtaining the final emotion class.

2.2. Emotional speech dataset annotation using LLMs

Fig. 1 shows the flow of emotional speech dataset annotation using LLMs. First, we extract the speech's acoustic features and text content. We use the transcription available in the dataset as recent automatic speech recognition (ASR) performance is very close to human transcribers and can be assumed to be the same. We then construct prompts as input to LLM using the transcriptions and the acoustic feature descriptors, which are then outputted to a single emotion class designated in the prompt. The example prompts are included in Fig. 1. We evaluated the annotation result using LLMs in different settings.

2.2.1. Single utterance and whole conversation

LLM showed capabilities on NLP-based tasks, such as summarizing long texts or dialogues. In other words, LLM is good at extracting context. Some studies in SER [21,22] show that processing SER in conversational sequences helps in improving SER performance. This is because the emotion of a particular utterance is affected by the emotion from the previous utterance. In short, the context of conversation contains emotional changes. In this study, we conducted annotations both on single utterances and whole or part of conversations.

2.2.2. Inclusion of acoustic feature descriptors in prompt

Acoustic features contain essential information for emotion recognition and should not be ignored. One challenge is how to incorporate acoustic features, which are numerical, to the prompt. LLMs may be able to process certain acoustic feature descriptors and show changes to the annotation result. To some extent, LLMs are also trained to process numerical values and classify if the value is high or low for certain metrics; for instance, a speaking rate of five syllables per second in English is considered fast, while two syllables per second is considered slow. However, there might be differences between the knowledge trained in LLMs and human perception, and the knowledge might not be available in less common languages. To alleviate this issue, we incorporate acoustic feature descriptors inspired by a previous study [19]. First, we extract the acoustic features, decide the threshold according to the data, and translate them into words, which are included as part of the prompt, as shown in Fig. 1. The thresholds are taken from the lowest 30% and the highest 30% of the ordered numerical value of the acoustic feature. Each acoustic feature is classified as low, normal, or high. In this study, we use several acoustic features considered essential to determine emotions: speaking rate, articulation rate, loudness, pitch, and intensity. Speaking rate is the number of syllables expressed per second excluding the silent parts; articulation rate is the speaking rate with silent parts included. Loudness is the amplitude of a particular utterance; pitch is how high or low is the speaking tone, and intensity is the average energy per utterance. As an additional note, the threshold for pitch is differentiated between male and female speakers.

Table 1. Dataset specifications

Dataset	IEMOCAP	
Speakers	5 male and 5 female	
Utterance length	1 − 19 s	
# of utterances	Happy	1636
	Sad	1084
	Neutral	1708
	Angry	1103

2.3. SER using label from annotation

We use the BLSTM-attention-based SER as a classifier, which adopts the base method of our previous study [6]. The SER receives acoustic features and text information as the input. The acoustic features and the encoder for the text information are the same as the ones used in our previous study. However, contrary to the previous study, we use the transcription instead of the ASR result as recent state-of-the-art ASR models have reached performance similar to that of human performance [23]. The acoustic features and encoded transcription (text features) are extracted using a separate BLSTM-attention module, which results in intermediate layer representations. These intermediate layer representations are then concatenated and fed to a dense network, resulting in the output class probabilities. To measure the SER performance, we conduct the experiments using the emotional speech data with some or all labels from the annotation result.

3. EXPERIMENTS

3.1. Dataset

To evaluate the effectiveness of the proposed method, we used the Interactive Emotional Dyadic Motion Capture (IEMOCAP) dataset [20], one of the benchmark datasets for emotion recognition. The IEMOCAP dataset consists of emotional speeches divided into five sessions, each containing one male and one female speaker. There are five male and five female speakers in the dataset. Each utterance corresponds to the transcriptions and is labeled as one of seven emotions (happy, sad, neutral, angry, excited, frustrated, and other). Similarly to previous works, we included the utterances labeled as excited to those labeled as happy, and we only used the data from four classes (happy, sad, neutral, and angry). We conducted five-fold cross-validation, in which four sessions were used as the training set, and the remaining one was used as the test set, ensuring speaker independence. The details about the dataset are shown in Table 1.

3.2. Annotation experiment settings

Our experiment uses GPT-3.5-turbo [1] for annotation experiments, which is the model behind the ChatGPT. The prompt is fed to GPT-3.5-turbo with a temperature of 0, max_token set to 500, and logit_bias for each emotion class to 10 to ensure a higher probability of the result being the selected emotion class. When the prompt answer is outside of the selected emotion class, we conduct another prompt to decide which is closest to the provided prompt answer. Due to the token restriction of each query in an LLM, we only include one type of acoustic feature for each prompt. We also limit the length of conversation segments to a maximum of 20 previous utterances from the input utterance for the experiment using conversation context. The final result of the prompt using acoustic feature descriptors is decided through the majority vote from each acoustic feature descriptor.

[1]https://platform.openai.com/docs/models/gpt-3-5

Table 2. Annotation performance comparison

Prompt type	Acoustic feature	UA (%)	WA(%)
Single utterance (without context)	None	50.4	50.8
	Speaking rate	48.7	49.6
	Articulation rate	48.8	49.5
	Loudness	51.7	52.1
	Pitch	50.1	51.3
	Intensity	47.2	48.3
	Majority vote	50.8	51.5
Conversation (with context)	None	65.1	64.2
	Speaking rate	71.1	70.2
	Articulation rate	72.6	71.6
	Loudness	71.1	70.1
	Pitch	71.2	70.4
	Intensity	70.8	70.2
	Majority vote	72.4	71.4
Human annotation [26]		69.0	70.0

Table 3. SER performance comparison on different amount of annotation data

LLM annotation data amount	UA (%)	WA(%)
100% LLM annotation	72.3	73.0
50% original	75.2	75.6
50% original 25% LLM annotation	76.6	76.9
50% original 50% LLM annotation	77.3	77.4
100% original	78.3	78.1

3.3. SER experiment settings

For the SER experiment, we use the base of our previous method [6], consisting of an acoustic feature extractor, a textual feature extractor, and an emotion classifier. The input mainly follows the previous study, except for using transcription instead of ASR results. We extracted a 33-dimensional feature consisting of 20-dimensional MFCCs, 12-dimensional CQT, and one-dimensional F0 for acoustic feature extraction. All of the acoustic features are extracted using Librosa [24]. For the textual features, we use the transcriptions provided in the dataset, encoded using BERT [25] pretrained using lower-case English texts. The pretrained BERT consists of 12 layers and 110M parameters, resulting in 768-dimensional textual features. The specifications for the SER method are the same as the one used in the previous study. We employed unweighted accuracy (UA) and weighted accuracy (WA) as evaluation metrics for SER.

Table 4. SER performance comparison to state-of-the-art methods

Method	UA (%)	WA(%)
Zou et al. [4]	72.7	71.6
Kang et al. [5]	75.4	76.0
Padi et al. [13]	75.8	76.1
Gao et al. [11]	76.1	74.9
Priyasad et al. [9]	76.8	77.3
Morais et al. [12]	77.1	76.6
Kim et al. [7]	77.7	77.4
Wu et al. [14]	78.4	77.5
Peng et al. [10]	81.4	80.3
LLM annotation result	72.4	71.4
SER, 100% LLM annotation	72.3	73.0
SER, 50% original 50% LLM annotation	**77.3**	**77.4**

3.4. Experiment result

3.4.1. Annotation experiment

The result of the annotation is shown in Table 2. Our experimental results show that using a single utterance for prompting without acoustic feature descriptors yields a UA and WA of 50.4% and 50.8%, respectively. In the single utterance prompting, although adding acoustic feature descriptors provides a context to how the text is spoken to LLM and can change the annotation outcome, it does not always improve annotation performance. For instance, adding loudness slightly improved the annotation, while adding the speaking rate decreased the performance. To resolve the performance fluctuations and replicate the human annotation procedure, we combine the result of prompting using acoustic feature descriptors and conduct a majority voting for the annotation labels. The majority voting use all results from each acoustic feature descriptor, resulting in a performance close to the best one from the single acoustic feature descriptor and better than the one without an acoustic feature descriptor.

On the other hand, using the conversation segment as the prompt improves the annotation performance, yielding UA and WA of 65.1% and 64.2%, respectively, increasing UA and WA by 14.7 and 13.4 percentage points, respectively. One possible reason for this improvement is that the emotion in a certain utterance is affected by the emotion from the previous utterances. Moreover, LLM was shown to be extremely capable of extracting context and incorporating it to influence the prompt result. The context extracted by LLM includes the context carried by acoustic feature descriptors' sequential changes. Compared to using only conversation context, the result of using both conversation context and acoustic feature descriptors is improved by around 5.5–7.5 percentage points in both UA and WA. This finding suggests that LLM also extracts context from the dynamics in the acoustic feature descriptors aside from context extracted from the content. Similar to the result on prompting using a single utterance, the performance of the majority vote system in the result using context information is close to the best performance from using conversation context and a single acoustic feature descriptor, with UA and WA of 72.4% and 71.4%, respectively. This performance is higher than the reported performance on human annotation [26], which is around 69% and 70% for UA and WA, respectively. Note that the human annotation performance in [26], in which the study relabeled the IEMOCAP dataset and decided the emotion label from the majority of five annotators, is different from the original labels. Considering the high cost and possible inconsistency of human annotators shown in the relabeling experiment, the annotation experiment using LLM would cost less while having the potential to be more reliable.

In using LLM to annotate emotional speech, we noticed several limitations that might result in performance fluctuations. First, LLM performance is sensitive to prompt changes, and therefore, formulating an effective prompt might be essential to obtain a result as close as possible to the desired class. Second, the annotation result changes based on the acoustic feature descriptors passed to the prompt; therefore, setting the threshold and words to decide the acoustic feature descriptors may affect the performance.

3.4.2. SER using LLM-annotated data

For the SER using LLM-annotated data, the result is shown in Table 3. Here, we adjust the amount of data using LLM annotated labels and the original dataset, with the percentage showing from the overall training data. First, as a base performance, using 100% original labels yields a UA and WA of 78.3% and 78.0%, respectively. On the other hand, using 100% labels from annotation yields lower performance, with UA of 72.3% and WA of 73.0%, similar to the performance of annotation using LLM shown in Table 2. The slight decline in SER performance using all annotated labels can be considered acceptable considering the annotation performance.

Second, we investigate a case when the annotation data is set as augmentation data. In these cases, we limit the data using original labels to 50% and adjust the remaining data with labels from annotations: no annotation data, 25% of all training data, and 50% of all training data. The result shows that adding annotation data as augmentation improves the UA and WA up to 77.3% and 77.4%, respectively, therefore opening the potential to use the data using labels from annotation for data augmentation. Our method using annotation output, SER using 100% LLM annotation result, and SER using 50% original and 50% LLM annotation result is also shown to have comparable performance to the state-of-the-art methods, as displayed in Table 4.

4. CONCLUSIONS

We proposed a method to annotate emotional speech data using LLM and incorporate the annotation result to evaluate the performance of SER. Our experiments suggest that including conversation context and the acoustic feature descriptors improves annotation performance, outperforming human annotators in performance and cost. Moreover, although the performance of SER using data annotated using LLM is less than that of original data, the performance is not too far behind that of state-of-the-art methods. In addition, we found that data annotated using LLM can be effectively used as additional augmented data, possibly improving the SER performance in low-resource conditions.

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